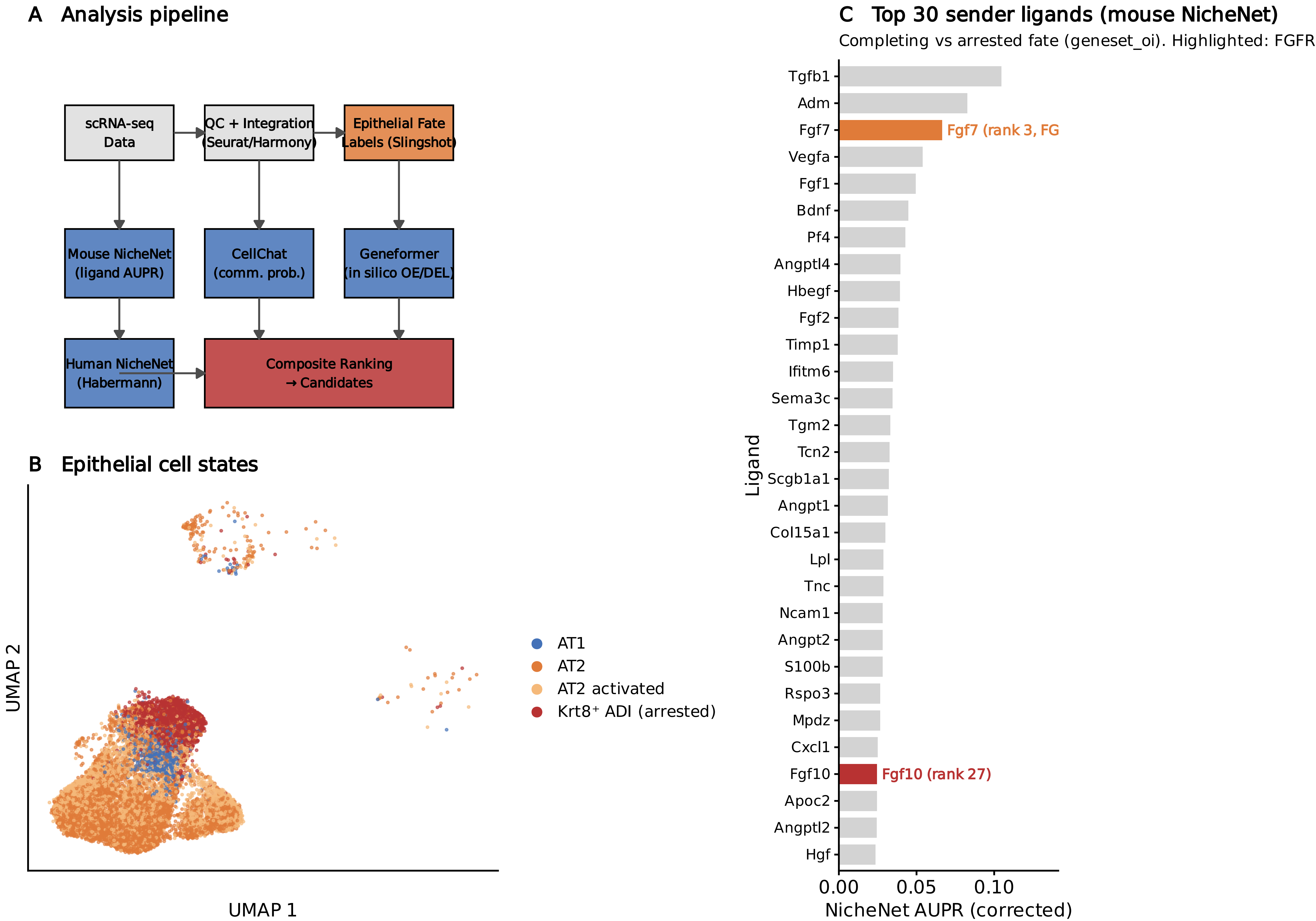
A Analysis pipeline
scRNA-seq Data
QC + Integration (Seurat/Harmony)
Epithelial Fate Labels (Slingshot)
Mouse NicheNet (ligand AUPR)
CellChat (comm. prob.)
Geneformer (in silico OE/DEL)
Human NicheNet (Habermann)
Composite Ranking → Candidates
B Epithelial cell states
UMAP 1
UMAP 2
AT1
AT2
AT2 activated
Krt8⁺ ADI (arrested)
C Top 30 sender ligands (mouse NicheNet)
Completing vs arrested fate (geneset_oi). Highlighted: FGFR
Tgfb1
Adm
Fgf7
Fgf7 (rank 3, FG
Vegfa
Fgf1
Bdnf
Pf4
Angptl4
Hbegf
Fgf2
Timp1
Ifitm6
Sema3c
Tgm2
Tcn2
Scgb1a1
Angpt1
Col15a1
Lpl
Tnc
Ncam1
Angpt2
S100b
Rspo3
Mpdz
Cxcl1
Fgf10
Fgf10 (rank 27)
Apoc2
Angptl2
Hgf
Ligand
0.00
0.05
0.10
NicheNet AUPR (corrected)

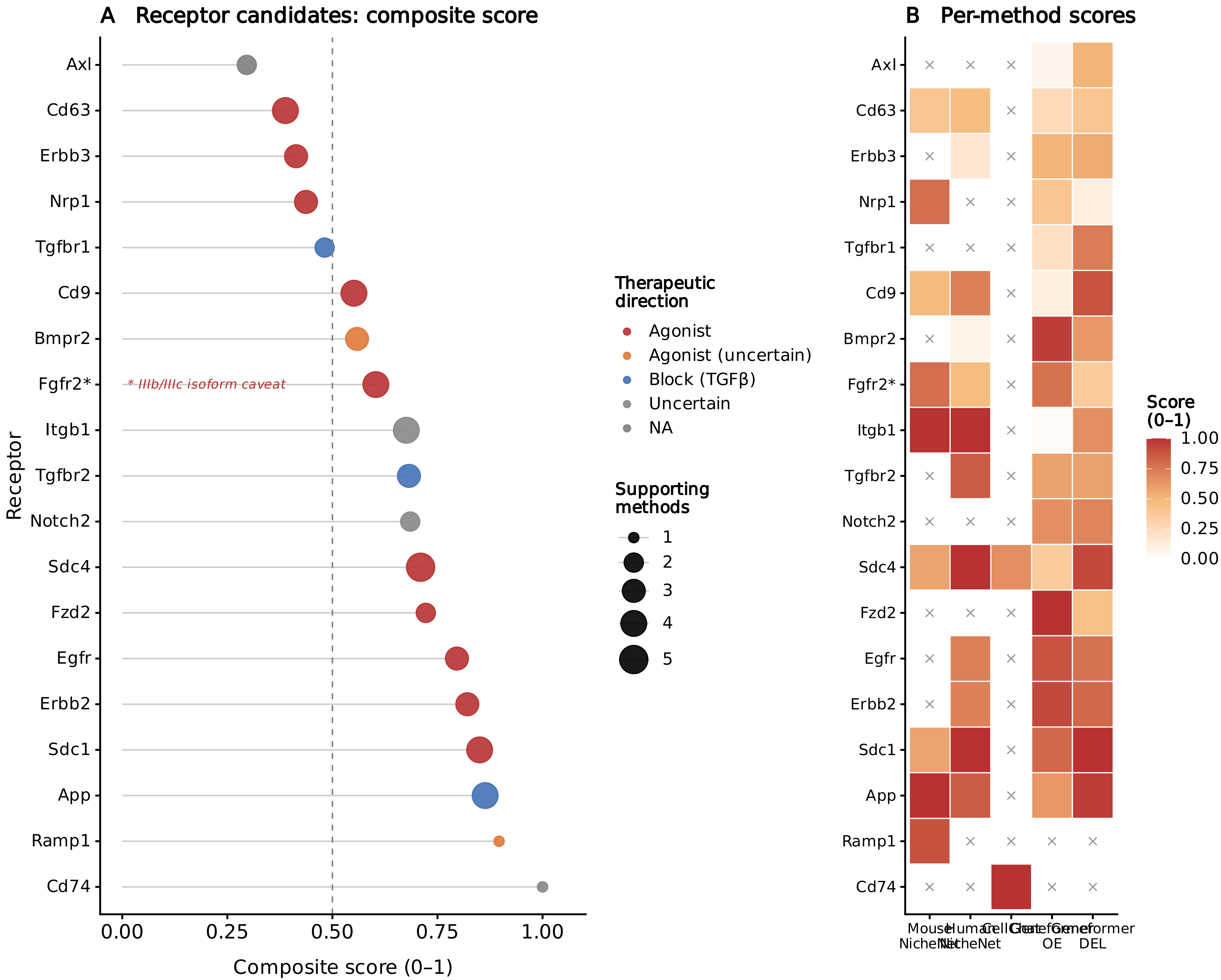
A Receptor candidates: composite score
B Per-method scores
Receptor
Axl
Cd63
Erbb3
Nrp1
Tgfbr1
Cd9
Bmpr2
Fgfr2*
Itgb1
Tgfbr2
Notch2
Sdc4
Fzd2
Egfr
Erbb2
Sdc1
App
Ramp1
Cd74
* IIIb/IIIc isoform caveat
0.00
0.25
0.50
0.75
1.00
Composite score (0–1)
Therapeutic direction
Agonist
Agonist (uncertain)
Block (TGFβ)
Uncertain
NA
Supporting methods
1
2
3
4
5
Score (0–1)
1.00
0.75
0.50
0.25
0.00
Mouse NicheNet
Human NicheNet
CellChat
Geneformer OE
Geneformer DEL

Figure 3 — FGF10 sender-side: supply is episodic in mouse and low in human IPF

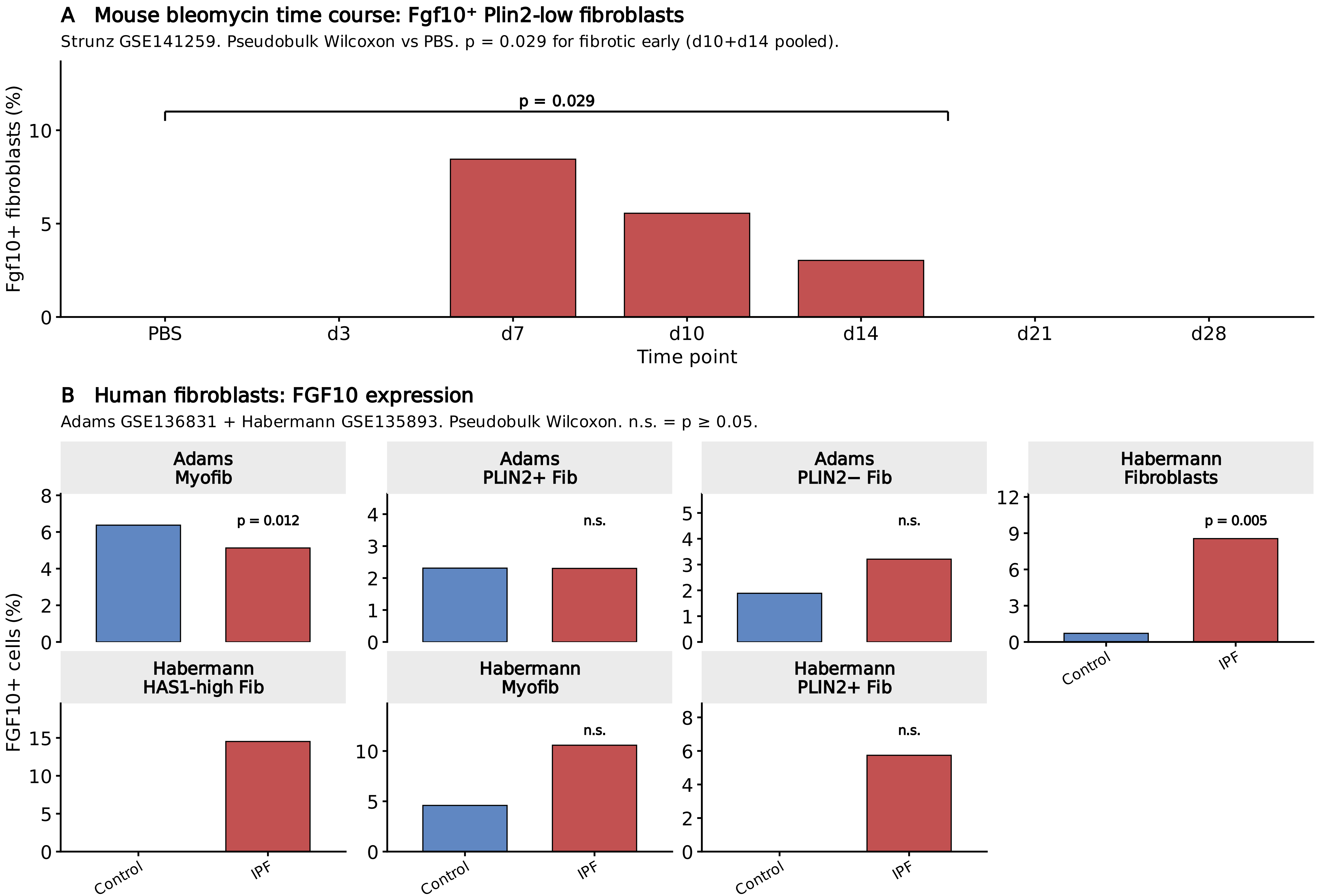

Figure 4 — FGFR2 receiver: expression maintained; co-expressed with arrest markers